

Text Summarization across Different Platforms

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Text Selected for Summarization:

Blockchain technology is a decentralized, distributed, and immutable digital ledger designed to record transactions across a network of computers, ensuring data integrity without a central authority. At its core, it functions as a chronological chain of data blocks, where each block contains a batch of transactions, a timestamp, and a unique cryptographic hash of the preceding block. This cryptographic linkage makes the ledger tamper-evident; altering any historical data would require recalculating all subsequent blocks, which is computationally infeasible.

The network operates on a peer-to-peer (P2P) architecture, where participants, known as nodes, maintain a synchronized copy of the entire ledger. To ensure data consistency across these distributed nodes, the system utilizes consensus mechanisms—such as Proof of Work (PoW) or Proof of Stake (PoS)—to validate new transactions and reach agreement on the state of the network. This decentralized validation removes the need for trusted intermediaries, effectively eliminating single points of failure and reducing risks associated with double-spending.

Beyond basic transaction storage, blockchain technology often incorporates smart contracts, which are self-executing scripts that automate actions when predefined conditions are met. This combination of immutability, decentralization, and transparency provides a secure infrastructure

for diverse applications, ranging from cryptocurrency to supply chain management and decentralized finance.

AI Platforms Used:

- ChatGPT (OpenAI)
- Gemini (Google)
- Copilot (Microsoft)

Execution:

Each platform received the same article and was prompted with each technique. Outputs and processing times were recorded for comparison.

Evaluation Criteria:

- Accuracy: Correctness in capturing the article's key points.
- Coherence: Logical sequencing and readability.
- Simplicity: Accessibility of language for undergraduate students.
- Speed: Time taken to generate the summary.
- User Experience (UX): Interface intuitiveness and ease of saving output.

Each criterion was scored on a scale from 1 (poor) to 5 (excellent).

Prompting Techniques Defined:

1. Zero-Shot Prompting:

Prompt:

- Summarize the text without giving any examples.

Responses generated by AI Tools:

⇒ ChatGPT:

Blockchain technology is a decentralized and distributed digital ledger that records transactions across a network of computers without relying on a central authority. It organizes data into sequential blocks linked through cryptographic hashes, making the ledger secure and resistant to tampering. A peer-to-peer network of nodes maintains synchronized copies of the ledger and uses consensus mechanisms to validate transactions and maintain consistency. The technology also supports smart contracts, enabling automated actions based on predefined conditions while providing a secure and transparent system for digital record-keeping.

⇒ Gemini:

Here is a summary of the key concepts from the text:

Blockchain Technology Overview

- **Core Concept:** It is a decentralized, immutable digital ledger that securely records transactions across a peer-to-peer network without relying on a central authority.
- **Data Security:** Information is structured in a chronological chain of cryptographically linked blocks. This design makes the system tamper-evident, as altering historical data is computationally unfeasible.
- **Network Operation:** Distributed nodes maintain a synchronized copy of the ledger by using consensus mechanisms to validate new transactions and agree on the network's state. This eliminates the need for trusted intermediaries and removes single points of failure.
- **Automation:** The technology utilizes smart contracts—self-executing scripts—to automatically perform actions when predefined conditions are met, creating a secure, transparent, and decentralized infrastructure.

⇒ Copilot:

Blockchain technology is a decentralized and immutable digital ledger that records transactions in a chronological chain of data blocks. Each block is cryptographically linked to the previous one, making alterations computationally impractical. The system operates on a peer-to-peer network where nodes maintain synchronized copies of the ledger, and consensus mechanisms ensure data consistency and validation. It also supports automated execution through smart contracts,

combining transparency, security, and decentralization to provide a reliable infrastructure for digital interactions.

2. Few-shot Prompting:

Prompt:

- Summarize the text with giving two or three examples.

Responses generated by AI Tools:

⇒ ChatGPT:

Blockchain technology is a decentralized and distributed digital ledger that records transactions across a network without a central authority. It stores data in linked blocks secured by cryptographic hashes, making the system transparent and tamper-resistant. Transactions are verified through consensus mechanisms and maintained by a network of nodes, ensuring reliability and security. It can also run smart contracts that automatically execute actions when certain conditions are met.

Examples of its applications include cryptocurrency transactions, supply chain tracking, and decentralized finance systems.

⇒ Gemini:

- Blockchain Technology Summary
- Core Concept: Blockchain is a decentralized, immutable digital ledger that securely records transactions across a peer-to-peer network, completely removing the need for a central authority.
- Security and Consensus: Data is structured in a chronological chain of cryptographically linked blocks. Distributed nodes maintain the ledger and use consensus mechanisms (such as Proof of Work or Proof of Stake) to validate new transactions, making the system tamper-evident and eliminating single points of failure.
- Automation: The network often utilizes smart contracts—self-executing scripts that automatically trigger actions when predefined conditions are met.

- Examples of Blockchain Applications As highlighted in the text, this combination of decentralization, immutability, and transparency provides a secure infrastructure for diverse uses, such as:

- Cryptocurrency
- Supply chain management
- Decentralized finance

⇒ Copilot:

Blockchain technology is a decentralized and immutable ledger that records transactions in linked blocks, secured by cryptographic hashes. It operates on a peer-to-peer network where nodes maintain synchronized copies, and consensus mechanisms like Proof of Work or Proof of Stake validate new entries. In addition to transaction recording, it supports smart contracts—self-executing scripts that automate actions. This combination of transparency, security, and decentralization enables applications such as cryptocurrency, supply chain management, and decentralized finance.

3. Chain Of Thought Prompt:

Prompt:

- Based on the above two summarization, give me the detailed summary of block chain Technology.

Responses generated by AI Tools:

⇒ ChatGPT:

Blockchain technology is a decentralized and distributed digital ledger that records transactions across a network of computers without the need for a central authority. Instead of relying on a single controlling organization, the system is maintained by multiple participants in a peer-to-peer network, ensuring transparency and reducing the risk of manipulation or failure. Each participant, known as a node, keeps a synchronized copy of the ledger so that the data remains consistent and accessible across the entire network.

The information in a blockchain is stored in blocks, which are groups of transactions. Each block contains a timestamp, transaction data, and a unique cryptographic hash that links it to the previous block. This linking forms a continuous chain of blocks. Because every block is connected to the one before it, any attempt to alter earlier data would require changing all the following blocks in the chain, making the system highly secure and resistant to tampering.

To maintain agreement among the distributed nodes, blockchain networks use consensus mechanisms such as Proof of Work (PoW) or Proof of Stake (PoS). These mechanisms allow the network to verify and validate transactions before they are added to the ledger. As a result, the system can operate securely without relying on trusted intermediaries and helps prevent issues like fraudulent transactions or double-spending.

Another important feature of blockchain technology is the use of smart contracts. Smart contracts are self-executing programs stored on the blockchain that automatically perform actions when specific conditions are met. This automation improves efficiency, reduces the need for manual verification, and increases reliability in digital transactions.

Due to its decentralization, transparency, and strong security features, blockchain technology is widely used in many fields. Common applications include cryptocurrency systems, where digital currencies are transferred securely; supply chain management, where product movement and authenticity can be tracked; and decentralized finance (DeFi) platforms, which enable financial services without traditional intermediaries. Overall, blockchain provides a reliable and secure framework for storing and managing digital information in a transparent and distributed environment.

RESULT

PROMPT TECHNIQUE	PLATFORM	ACCURACY	COHERENCE	SIMPLICITY	SPEED	UX
ZERO-SHOT PROMPTING	CHATGPT	4.2	4.5	4	5	5
	GEMINI	4.7	4.6	3	3.7	4.5
	COPILOT	3	3.9	4.2	4.1	4
FEW-SHOT PROMPTING	CHATGPT	4	4.3	4	5	5
	GEMINI	5	4.2	3	4	4.9
	COPILOT	3.3	4	4	4.3	3
CHAIN-OF-THOUGHT PROMPTING	CHATGPT	4	4.7	4	5	4
	GEMINI	4.5	4	4.2	4	5
	COPILOT	4	3.9	4	4.3	3
ROLE-BASED PROMPTING	CHATGPT	3.8	4.5	4	5	4.7
	GEMINI	4.5	4.2	4.2	4	5
	COPILOT	3.9	4	4	4	4